

⊕ SUBNET MAGAZINE

OP-ED · Issue 008, First principles

What is Bittensor? An Open Letter and Reflection

A reflection on how to articulate Bittensor, "The Great Mediator", and the case for incentivized learning as machine learning's fourth paradigm.



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and Reflection

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When I ask the question, "What is Bittensor?", I hear many artfully defined explanations and perspectives. For example, RBarrySilbert has said "Bittensor is a decentralized network on which AI applications are being built that are permissionless and decentralized in this ingenious incentive-based system for anybody to build on using subnets." @const_reborn has stated that Bittensor's subnets are micro-directed incentive designs, each specific to a certain commodity. RJosephJacks has said that he thinks Bittensor is reinforcement learning over any economic activity, intrinsically incentivized and proven at scale in production across dozens of computing form factors. @markjeffrey has said that open-source AI (RBittensor) will indeed drink the milkshake of companies like RAnthropicAI when referring to a particular subnet in the ecosystem (@ridges_ai) outperforming billion-dollar competitors on certain important benchmarks.

For many of us, Bittensor is difficult to understand, and for those who understand it, sometimes it can be difficult to explain. So how do we articulate Bittensor? How do we understand what Bittensor is? This is a complex topic as Bittensor is an emerging technology and philosophy, whose architectural attributes are several opposing parts that equate to its whole.

As I study Bittensor and reflect on what the many great minds in the space have said about the ecosystem, I have begun to try and construct my own understanding of what Bittensor is. This short article is my attempt at

that endeavor. In my lowly, humble, and simple opinion, I believe Bittensor to be The Great Mediator, the meeting point between machine intelligence, blockchain, incentivization, open source software, global compute, science, sound economics/tokenomics, etc., all in sync while at the same time being comprised of opposing and contrasting characteristics—a multi-disciplinary and inter-disciplinary technology that shares a unique distribution of value and interdependency through multiplicity. In short, Bittensor is a kind of transcendent intelligence network.

In many senses, I see Bittensor as pioneering the fourth paradigm in machine learning, which has traditionally been framed around three primary paradigms, each addressing different ways to extract knowledge from datasets or environments.

These traditional paradigms are:

1. Supervised Learning: This is where models learn from labeled datasets, or as RRichardSSutton explains, "supervised learning is learning from a training set of labeled examples provided by a knowledgeable external supervisor." It's like having a guide pointing out right and wrong to shape the model's predictions.
2. Unsupervised Learning: This is basically about finding structure hidden in collections of unlabeled data, where the model uncovers patterns, clusters, or relationships on its own without any predefined labels or oversight.
3. Reinforcement Learning: From my understanding, this is about trying to maximize some type of reward signal, and one of the most difficult challenges—which I don't think the other two suffer from—is a sort of trade-off between exploration and exploitation. The agent has to strive for the maximum amount of reward utilizing known paths (exploitation) and make decisions to take unknown paths (exploration) to discover new ways to gain rewards; it's a type of goal-seeking agent.
4. Incentivized Learning: I believe Bittensor is pioneering a shift in machine learning, leading the fourth paradigm in the machine computational approach I would like to call "incentivized learning" or decentralized incentivized learning. This approach leverages blockchain-based rewards to coordinate global, peer-to-peer machine learning contributions, aligning individual optimizations with collective intelligence production. In a sense, it's building on reinforcement learning principles at a global network scale, treating participants as agents in a multi-agent system optimizing for shared reward. Participants "mine" intelligence, transforming machine learning into a commoditized, global resource.

Those who truly believe in the Bittensor Way are on the path of the four intellectual journeys.

The pilgrims among us are striving towards creation and Intelligence. The first journey is from the creation to the Intelligence. The second journey is through the intelligence in the Intelligence. The third journey is opposite of the first because it is from the Intelligence to the creation. The fourth journey is the opposite of the second because it is by the Intelligence in the creation.

I will continue to reflect on this matter as I do not do justice to the protocol.

In closing I'll leave you with the words of Barry Silbert: "If you asked five people to define Bittensor, you will get 10 different explanations". Its definition is hard to explain and its concept is difficult to capture. It is

intelligence in motion.

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Simplifying Bittensor subnet ecosystem through accessible, insightful articles. #TAO